LAB 7 - FUZZ TESTING

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Terminal-style reproduction of output from the uploaded PDF.

1. Initial application output

Run on a locally prepared baseline with missing input checks.

▸ **Windows PowerShell • Terminal output**

```
PS> C:\Program Files\Python311\python.exe processor.py
Parsed integers: [2, 4, 6]
Normalized username: transit_user-7
Average: 20.0
```

2. Hypothesis tests before fixes

The baseline produces two failures on non-string inputs.

▸ **Windows PowerShell • Terminal output**

```
PS> C:\Program Files\Python311\python.exe -m pytest -q --tb=short test_processor.py
.......FF                                                                [100%]
================================== FAILURES ===================================
_________________ test_parse_int_list_rejects_non_string_inputs _________________
test_processor.py:67: in test_parse_int_list_rejects_non_string_inputs
    @given(st.one_of(st.none(), st.integers(), st.floats(), st.lists(st.integers())))
                   ^^^
test_processor.py:70: in test_parse_int_list_rejects_non_string_inputs
    parse_int_list(value)
processor.py:7: in parse_int_list
    return value.split(",")
           ^^^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'split'
E   Failing test case: test_parse_int_list_rejects_non_string_inputs(
E       value=None,
E   )
_____________ test_normalize_username_rejects_non_string_inputs _____________
test_processor.py:74: in test_normalize_username_rejects_non_string_inputs
    @given(st.one_of(st.none(), st.integers(), st.floats(), st.lists(st.integers())))
                   ^^^
test_processor.py:77: in test_normalize_username_rejects_non_string_inputs
    normalize_username(value)
processor.py:30: in normalize_username
    return value.strip().lower()
           ^^^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'strip'
E   Failing test case: test_normalize_username_rejects_non_string_inputs(
E       value=None,
E   )
=========================== short test summary info ===========================
FAILED test_processor.py::test_parse_int_list_rejects_non_string_inputs - Att...
FAILED test_processor.py::test_normalize_username_rejects_non_string_inputs
2 failed, 7 passed in 3.28s
```

3. Fixed application output

Run on the processor included with this submission.

▸ **Windows PowerShell • Terminal output**

```
PS> C:\Program Files\Python311\python.exe processor.py
Parsed integers: [2, 4, 6]
Normalized username: transit_user-7
Average: 20.0
```

4. Hypothesis tests after fixes

All nine tests passed.

▸ **Windows PowerShell • Terminal output**

```
PS> python -m pytest -q --tb=short test_processor.py
.........                                                [100%]
9 passed in 3.09s
```